TERRAGUARD AI

THREE INCIDENTS. ONE CREW.

A hotspot is not a decision.

Evidence-aware wildfire incident prioritization

by Team: Dream Benders

DORMATHON 2026 · TRACK 1



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The cost of a late decision is real.

The challenge: limited people, many uncertain satellite alerts.

INDONESIA, 2015



US$16.1 billion

estimated economic losses

2.6 million

hectares burned

MANY INCIDENTS

- INCIDENT A
- INCIDENT B
- INCIDENT C

→

ONE AVAILABLE CREW

01

TARGET USER | Forest protection coordinator — many incidents, one available crew

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Detect. Categorize. Prioritize. Explain.

One end-to-end decision workflow—not another hotspot map.

01 DETECT

NASA FIRMS hotspots + weather

Group observations into incidents.

02 CATEGORIZE

Exposure • persistent activity • uncertain

Flag weak or unfamiliar evidence.

03 PRIORITIZE

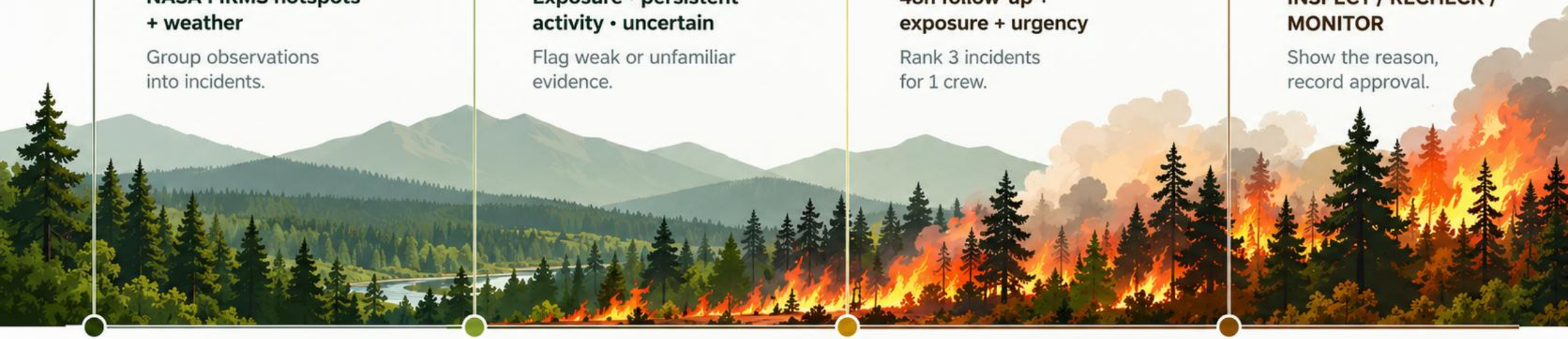
48h follow-up + exposure + urgency

Rank 3 incidents for 1 crew.

04 EXPLAIN & ACT

INSPECT / RECHECK / MONITOR

Show the reason, record approval.



OBSERVATIONS — INCIDENT CONTEXT — RANKED RESPONSE — HUMAN APPROVAL

TARGET USER

Forest protection coordinator

Makes one better-informed decision when only one field team is available.

48h = qualifying follow-up satellite detection window, not a fire-spread forecast.

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Expected Impact

The intended benefit of each feature

Satellite + weather
Combines scattered evidence in one workflow.

48-hour follow-up prediction
Focuses attention on likely repeat detections.

Explainable prioritization
Makes allocation more transparent and consistent.

Evidence-aware triage
Supports cautious action when evidence is weak.

Earth View + map layers
Improves geographic understanding.

Dated imagery comparison
Reviews visual and vegetation changes over time.

Distance + area measurement
Supports spatial assessment, not burned-area estimates.

3D response testing
Explores response trade-offs before action.

Unattended incidents
Makes the opportunity cost of assignment visible.

Human approval + records
Preserves responsibility and a reviewable audit trail.

Grounded AI briefings
Communicates decisions without invented numbers.

Local record fallback
Preserves records during remote-storage outages.

Expected operational benefits, not measured field outcomes.



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A real decision loop, not just a score.

Command map → 3D response test → human decision

REAL COMMAND MAP • HISTORICAL REPLAY



3D RESPONSE TEST • HYPOTHETICAL



1 SELECT & PREDICT
Choose a historical incident.

2 TEST THE RESPONSE
Replay matched scenarios in 3D.

3 APPROVE & LOG
Save an inspection or recheck.

3D visuals are hypothetical response scenarios, not a validated wildfire forecast.

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Technology Stack

The technologies identified in the reviewed source

Frontend	Next.js 16.4.0, React 19.1.0, TypeScript, Tailwind CSS 4
3D visualization	Three.js, React Three Fiber, Drei
Mapping	MapLibre GL with the Esri World Imagery basemap
UI and state	Zustand, Recharts, Lucide React
Backend	Python, FastAPI, Uvicorn, Pydantic
Data and ML	NumPy, pandas, scikit-learn, PyArrow, joblib; Random Forest
Simulation	Python/NumPy cellular automaton with R3F visualization
Decision records	SQLite locally; Supabase when configured

Versions are repository declarations, not verified installed versions.



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Thank you

By Team: Dream Benders

